

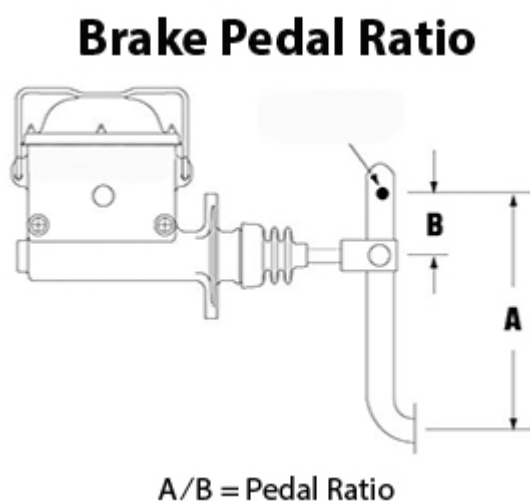
What is brake pedal ratio?

Pedal ratio is an important but often overlooked part of your braking system. It tells you how the force you apply to the pedal is multiplied and transferred to the [Master Cylinder](#). It also affects how far the pedal travels.

Why is it important?

The average person can press on the brake pedal with about 70 lbs. of force. However, to stop a moving vehicle, disc brakes require 800-1,200 psi at the calipers. The force you put on the pedal is amplified by all the parts of the system. The pedal ratio is one thing that helps.

How is it calculated?



Measure the distance from the pedal's pivot point to the center of the pedal pad.

This is measurement "A" in the diagram.

Measure the distance from the pedal's pivot point to the center of the master cylinder pushrod.

This is measurement "B" in the diagram.

“A” divided by “B” equals the pedal ratio.

For Example

Assume the following:

Measurement “A” = 12 in.

Measurement “B” = 3 in.

12 divided by 3 = 4. In this case, the pedal ratio is 4:1.

How does it affect performance?

Using the numbers from the examples above shows how the force is multiplied:

70 lbs. of force applied to the pedal

A 4:1 pedal ratio

Equals 280 lbs. of force applied to the master cylinder pushrod

For power brake systems, the [Brake Booster](#) increases the force even more. This makes stopping easier and driving more enjoyable.

Also, take note of the pedal travel:

A 4:1 pedal ratio

A master cylinder with a 1.10 in. stroke

The pedal needs to travel 4.40 in.

It's important that the master cylinder bottoms out before the pedal hits the floor. This ensures the master cylinder moves enough fluid to stop the vehicle.

What should the pedal ratio be?

Pedal ratio is not something that is easily changed. However, it is something to consider during a custom build. It also comes into play when converting from manual to power brakes or vice versa.

The right pedal ratio depends on the brake system design. The pedal ratio needs to work with:

Disc vs. Drum Brakes

Master Cylinder Diameter

Wheel Cylinder Diameter

Caliper Piston Diameter

The chart below lists some basic guidelines. With some fabrication, your pedal ratio can be fine-tuned to meet your needs.

Brake System Type	Recommended Pedal Ratio
Manual	5:1 - 7:1
Power Assisted	4:1 - 5:1

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